

A PD4 group without an aspherical manifold model

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Abstract

We construct a finitely presented integral Poincaré duality group of dimension four that has a finite classifying space but is not the fundamental group of any closed aspherical topological four-manifold. This gives a negative answer to Wall’s manifold-realization question in dimension four.

1 Introduction

A closed aspherical manifold is determined up to homotopy by its fundamental group. Poincaré duality therefore supplies a necessary algebraic condition on that group. The manifold-realization question asks whether this condition, together with the appropriate finiteness assumptions, is sufficient. Here the category matters: a finite Poincaré complex, a homology manifold, and a smooth manifold are different possible realizations. We address closed *topological* manifolds in the same dimension as the duality group.

An oriented integral Poincaré duality group G of dimension n , or *oriented integral PD $_n$ group*, has a finite resolution of the trivial $\mathbb{Z}G$ -module $\mathbb{Z}$ by finitely generated projective modules; its cohomology with coefficients in $\mathbb{Z}G$ vanishes outside degree n and is $\mathbb{Z}$ in degree n , with trivial right G -action. A finite classifying space is a finite CW complex with fundamental group G and contractible universal cover. Such a space provides a finite free resolution, but its existence alone does not provide a manifold model.

Wall asked whether every integral Poincaré duality group is the fundamental group of a closed aspherical manifold [21, Problem G2]. The finiteness condition requires care. Davis used reflection groups to turn the finiteness examples of Bestvina and Brady into integral PD $_n$ groups that are not finitely presented, for every $n \geq 4$ [6, Theorem C]. Since a closed manifold has finitely presented fundamental group, these examples answer the unqualified question. The remaining manifold-realization question asks the same thing for *finitely presented* integral Poincaré duality groups; it is Question 3.4 in Davis’s survey [7] and Conjecture 7.29 in the cited version of Lück’s survey [14]. Kielak’s 2026 survey records that no counterexamples to this formulation were known [13, Section 4.1].

The coefficient ring is equally important. Fowler constructed torsion-free finitely presented rational Poincaré duality groups that cannot be realized by closed aspherical ANR rational homology manifolds [8, Theorem 2.1]. His examples have no finite classifying space. The example below instead satisfies integral duality and already has a finite four-dimensional classifying space; its obstruction concerns the passage from that Poincaré complex to a closed topological manifold.

Theorem 1.1. *There is a finitely presented group G with a finite, oriented, four-dimensional aspherical Poincaré complex Q such that*

$$H^i(G; \mathbb{Z}G) = 0 \quad (i \neq 4), \quad H^4(G; \mathbb{Z}G) \cong \mathbb{Z}$$

with trivial right G -action on the last group, but no closed aspherical topological four-manifold has fundamental group G . In particular the trivial left $\mathbb{Z}G$ -module $\mathbb{Z}$ has a finite resolution by finitely generated free modules.

Theorem 1.1 refutes the assertion quantified simultaneously over all dimensions $n \geq 3$, without asserting a counterexample in every dimension. The orientation qualification causes no additional issue: the orientation character of any hypothetical aspherical manifold with group G would equal the right action on its dualizing module and hence would be trivial.

1.1 The source of the obstruction

Our starting point is the marked geometric and tensor obstruction in *A marked tensor obstruction to four-dimensional disk embedding* [16]. Its geometric package records a finite family of based loop relations, including independently chosen conjugating paths, together with algebraic dual spheres. Its algebraic package turns a compatible cut configuration into potentials and exact identities over a ring with nilpotent parameters. A finite tensor theorem excludes those identities. Section 2.1 recalls the complete interface, including the mixed nilpotent terms and the memberships in actual gradient ideals. The marked geometry, algebraicity, formal comparison, and finite tensor theorem are inputs from the companion; the transfer to arbitrary marked topological fillings is proved here.

The argument has two principal geometric ingredients beyond that input. First, we construct a smooth chamber D with boundary S and a specified hyperbolic basis for its entire middle homology. Attaching three-cells along that basis gives a marked graph-type Poincaré pair (P, S) , but its marked boundary admits no topological graph-type filling. Here $H = \pi_1 D$ is free and BH denotes a finite graph with fundamental group H ; the space P is homotopy equivalent to BH . The marking is a specified map $S \rightarrow BH$. A *marked graph-type filling* is a compact topological four-manifold with identified boundary S and a homotopy equivalence to BH extending that map up to homotopy. This requirement retains information about the boundary loops throughout the construction. The extension of the cut obstruction to topological fillings uses smoothing off isolated points and a differential-form complex that computes the cohomology *with those points filled*. Second, we show that a hypothetical manifold model for a reflection of P would produce exactly such a forbidden filling. This requires a stable product with a specified sphere marking, followed by a destabilization argument. An unmarked stable homeomorphism would not suffice.

The chamber obstruction is a relative manifold-realization statement: the manifold must realize the homotopy type of the given Poincaré pair relative to its identified boundary. Davis and Hillman prove such relative realization for an aspherical Poincaré pair with nonempty closed manifold boundary in dimensions at least five, assuming the K - and L -theoretic Farrell–Jones conjectures, and in dimension four for elementary amenable fundamental groups [3, Theorem D]. Our chamber has nonabelian free fundamental group, and its marked filling obstruction concerns the four-dimensional case outside that elementary amenable hypothesis. Reflection converts this relative obstruction into the closed manifold-realization question of Theorem 1.1.

1.2 Proof architecture

The reflection method developed by Davis [5, Sections 13 and 15] provides the framework for closing the chamber. Our construction closes the pair to an aspherical Poincaré complex Q and reflects D to a closed manifold $N \rightarrow Q$. A support-tree construction gives the relevant right-angled Coxeter extension of the free chamber group a proper cocompact CAT(0) model. This auxiliary model supplies assembly information; the chamber development retains the four-dimensional

topology and its sphere classes. The CAT(0) action places $G = \pi_1 Q$ among the groups to which the Farrell–Jones theorems apply [1, 22]. If Q had a manifold model M , the four-dimensional surgery result of Kasprowski–Land [11], followed by a stable product theorem, would make all the prescribed hyperbolic pairs of a stabilization of N geometrically standard.

Lift these finite marked data to the reflection development. In a large finite union of chambers, cap every boundary summand except the central one. Remove standard pairs labelled by a large inner ball. The remaining sphere problems are far from the central chamber, where radial directions in the Davis complex make every bounded local construction small. Grope separation follows Freedman–Quinn’s finite-cover strategy [9, Section 2.10.2] and groups the upper branches over finitely many contractible direction patches. The essential requirement is that an entire resulting loop system stays over one such patch, even when it passes through many intersections. This replaces precisely the goodness-dependent null-loop step of Powell–Ray–Teichner [17], and not their theorem’s group hypothesis by fiat. The subsequent disk construction is uncontrolled and may range through the whole compact manifold. Surgery then gives the forbidden filling of S .

Section 2 constructs the marked chamber and proves its filling obstruction. Section 3 reflects it to obtain Q , verifies integral duality, and supplies the two geometric models used later. Section 4 uses a hypothetical manifold model to realize every prescribed sphere pair after stabilization. Section 5 caps a finite chamber union, removes the inner standard pairs, and uses direction control to remove the remaining middle homology. Its output is the marked filling excluded by Section 2.

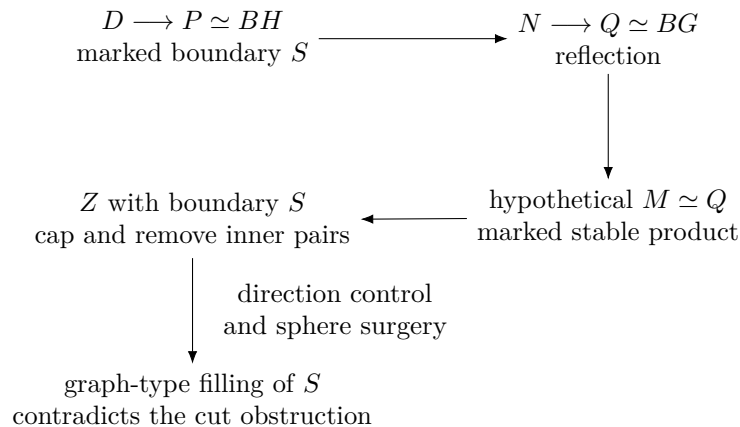


Figure 1: The two uses of the chamber. Its middle spheres permit the reflection and stable-surgery construction; its marked boundary forbids the filling obtained at the end.

1.3 Conventions

All manifolds in the surgery argument are oriented and all unqualified homology groups have integral coefficients. For a connected space with fundamental group Γ , regular coefficients $\mathbb{Z}\Gamma$ mean the homology of its universal cover. We fix lifts and paths to a basepoint when recording sphere classes. The involution on an oriented group ring is $\overline{\sum n_g g} = \sum n_g g^{-1}$. A hyperbolic plane over $\mathbb{Z}\Gamma$ has matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. For immersed sphere representatives, the phrase *framed hyperbolic system* also includes vanishing Wall self-intersection obstructions.

All stabilizations are interior connected sums with $S^2 \times S^2$ and leave boundary markings unchanged. Specific references to Freedman–Quinn [9] use the January 2013 reformatted edition’s numbering.

2 A chamber with no marked graph filling

The obstruction will first be placed in a compact chamber. Its boundary marking is essential: the obstruction does not merely assert that some disk problem has no solution.

Proposition 2.1 (The marked chamber). *There are a compact connected oriented smooth four-manifold D , its connected boundary S , a finitely generated free group H , and a map $c : D \rightarrow BH$ with the following properties.*

1. *The map c induces an isomorphism on fundamental groups, and D has the homotopy type of a finite graph with $2k$ two-spheres attached at its basepoint. Its module $\pi_2 D$ has a specified hyperbolic basis over $\mathbb{Z}H$, represented by framed immersed spheres with vanishing quadratic self-intersection obstructions.*
2. *Attach a three-cell to each basis sphere, and denote the resulting space by P . Then (P, S) is an oriented Poincaré pair of formal dimension four, and $P \simeq BH$. The boundary marking is $c|_S : S \rightarrow BH$.*
3. *There is no compact oriented topological four-manifold D' with boundary identified with S and a homotopy equivalence $c' : D' \rightarrow BH$ whose restriction to S is homotopic to $c|_S$.*

Here and below, sphere classes and intersections use fixed whiskers and deck translations. A hyperbolic pair has zero squares, mutual product 1, and zero framing and Wall self-intersection obstructions. We prove the proposition without using an embedded surgery on D .

2.1 The geometric and tensor inputs

We use the marked geometric construction of [16, Section 2], rather than only its main disk-nonembedding theorem. The marked bodies and exterior duals are supplied by its Lemmas 2.3 and 2.4. Its Proposition 2.2 and Lemmas 2.5–2.6 give the cut consequences under hypothetical disk replacements; those conditional conclusions are distinct from the marked geometric construction itself. We recall the precise data needed here. Fix $m \geq 5$ and $p \geq 9$. Start with a four-dimensional one-handlebody on $2m + p$ handles, attach cancelling two-handles to the first $2m$ handles with product framings, and plumb these two-handles in m pairs, interchanging the two coordinate directions at every plumbing. The resulting smooth oriented manifold X has graph homotopy type. Its cover associated to the plumbing quotient $\pi_1 X \rightarrow F_m$ is a tree of blocks, each a one-handlebody on p extra generators; neighboring blocks meet in a product four-ball. The exposed boundary of a block is

$$K_v = \#^p(S^1 \times S^2) \setminus \text{int } \nu(L_{2m}),$$

where L_{2m} is an unlink in a ball. Its port meridians $x_1, \dots, x_{2m}$ and extra loops $l_1, \dots, l_p$ form a free basis. At each join the meridian on one side is the longitude on the other.

Write $\mathcal{C} = \{1, l_1, \dots, l_p\}$. The marked-body construction installs a disjoint two-stage grope body for every ordered triple of distinct ports and every independent choice

$$X_a = c_a x_{r_a}^{\delta_a} c_a^{-1}, \quad c_a \in \mathcal{C}, \quad \delta_a \in \{1, -1\}, \quad a = 1, 2, 3.$$

Its lower surface is a punctured torus: one branch has a single cap, and the other carries the upper punctured torus with its two caps. All tip tracks, conjugating paths, and product framings are specified. The initial boundary word R satisfies

$$R|_{X_a=1} = 1 \quad (a = 1, 2, 3), \quad R \equiv [X_1, [X_2, X_3]] \pmod{\Gamma_4 F(X_1, X_2, X_3)}. \quad (1)$$

Internal changes of the marked paths are allowed by these identities; one must retain the entire finite family, not replace it by one unconjugated commutator.

Let $F_1, \dots, F_k$ be the lower punctured tori, with symplectic bases (a_j, b_j) , where a_j is the single-cap branch. Delete their open product tubes, including the proper edge collars, to obtain E . Lemma 2.4 (Algebraic duals in the exterior) of [16] supplies the truncated single caps f_j and framed spheres g_j in E such that

$$\lambda(f_i, g_j) = \delta_{ij}, \quad \lambda(g_i, g_j) = 0, \quad \tilde{\mu}(g_j) = 0 \quad \text{over } \mathbb{Z}[\pi_1 E]. \quad (2)$$

The framing on f_j is the surface-compression framing. The g_j are compressed normal-circle tori over the b_j curves; their individual representatives have trivial normal bundle. Equation (2) is a group-ring statement before any meridians are killed.

We also recall exactly how the cut obstruction is used. Suppose that a compact smooth oriented four-manifold with the marked graph type of the parity cover of X has two connected vertex pieces A, B , with outside faces K_A, K_B , and $2m$ connected three-dimensional cuts Y_i , each with its prescribed coordinate torus as its entire boundary. Orient Y_i as a face of A , so its orientation as a face of B is opposite. Suppose the differences of restrictions give an isomorphism in degree two and, in degree one, an isomorphism after the extra-loop evaluations are set to zero. Suppose further that precisely m joins are active: at an active join the surviving coordinate meridian is at A , whereas the other coordinate is the meridian at B . Require every marked law (1) in its respective vertex group, and the free ambient background in which the port meridians are trivial and the named extra letters can be assigned independently.

These data cannot exist. Indeed, Proposition 6.1 (Vertex data) of [16] gives a characteristic-zero field K (one may take $\mathbb{C}((\sigma))$), separate coordinate blocks h_i of total dimension N , and algebraic power-series germs $F_i(h_i)$ and $b_i(h_i)$, with b_i indexed by the active set I , $|I| = m$. The germs are normalized by $F_i(0) = b_i(0) = 0$. Set

$$T = K[t_i : i \in I] / (t_i^2 : i \in I), \quad S_0 = \sum_i F_i, \quad S_t = S_0 + \sum_{i \in I} t_i b_i. \quad (3)$$

There are formal parametrizations

$$H_L(x, t) \in (x, t)T[[x]]^N, \quad H_R(y) \in (y)K[[y]]^N,$$

where x has d coordinates and y has $N - d$ coordinates. Their central tangent maps $\partial_x H_L(0, 0)$ and $\partial_y H_R(0)$ are injective with complementary images in K^N ; nilpotent constant displacements of H_L are allowed. They satisfy

$$S_t(H_L) = 0, \quad S_0(H_R) = 0. \quad (4)$$

For every three distinct active indices i, j, k , they satisfy

$$t_i t_j t_k \in ((\partial_{h_\nu} S_t)(H_L) : \nu = 1, \dots, N) \subset T[[x]], \quad (5)$$

$$(b_i b_j b_k)(H_R) \in ((\partial_{h_\nu} S_0)(H_R) : \nu = 1, \dots, N) \subset K[[y]]. \quad (6)$$

Theorem 3.1 (Finite tensor obstruction) of [16] rules out (3)–(6) for $m \geq 5$. In particular, the memberships are in the actual gradient ideals, not their radicals, and T retains all mixed square-free monomials. The independent conjugators in the full law list are what supply all these triple identities. These two source results, together with the marked edge potentials, algebraicity and formal comparison of [16, Proposition 4.5, Theorem 5.1 and Proposition 5.2], are the tensor obstruction used here.

2.2 The exterior with its side marking

The imported geometry supplies the surfaces and their algebraic duals. We now use those data to construct D , keeping track of the actual tube-side maps. This marking will identify the boundary of a hypothetical filling with the loop data needed for the tensor contradiction.

We make one placement refinement of the marked-body construction. Place its longitudinal tip annuli on disjoint parallels outside sufficiently thin attaching solid tori in the initial one-handlebody boundary, and put the finitely many connectors in their complement. This is possible by the marked handlebody model: the bodies are boundary joins of the separated annular collars. The complement of the attaching solid tori survives in the boundary after the two-handles are attached; the plumbing ports are on the new belt-handle portions. Thus the unpushed lower surfaces lie in the final exposed boundary of X . The independent ribbon twists matching the cap framings are made in the separated parallel tubes. They do not pass a body across a plumbing join. The returning tip collars alone are lengthened back to product parallels on the attaching regions and then into the cores. The construction and intersection calculation of (2) are unchanged.

Consequently the F_j are proper inward pushes of disjoint boundary surfaces, with returning collars at their initial boundary curves. Moreover $F_j \rightarrow X$ is null-homotopic. The three tip loops bound the corresponding core caps in X ; hence the bottom a_j and b_j loops die in $\pi_1 X$. Since X has graph homotopy type and F_j retracts to a graph, this proves the assertion about the whole surface map.

Lemma 2.2 (The marked exterior model). *With the surface-product trivializations, the exterior and its tube-side maps have the homotopy-pushout model*

$$E \simeq X \cup \coprod_j F_j \coprod_j (F_j \times S^1),$$

where each gluing map to $F_j \times S^1$ is the inward section. The displayed $F_j \times S^1$ maps represent the actual tube sides.

Proof. It is enough to describe one surface; the constructions have disjoint collars. Write $B = \partial X$ and $K = \partial F$. Before rounding corners, put the pushed surface at a horizontal collar level, with vertical walls along K returning to the boundary. The inner part of the exterior has homotopy type X , the outer part has homotopy type $B \setminus K$, and their overlap has homotopy type $B \setminus F$. Product slices, thickened by a variable positive width off the deleted surface, give this homotopy-pushout decomposition.

In $B \setminus K$, take a two-sided neighborhood of $F^\circ = F \setminus K$ whose width decreases to zero near K . Relative to $B \setminus F$, it is a bridge over F° , joining the two sides of the cut. Replace F° in this diagram by a slightly shrunk copy F_0 ; this is an objectwise homotopy equivalence. The actual deleted surface and K are unchanged. In the rectangular normal coordinates to its horizontal copy, split the tube circle over F_0 into its outer and inward semicircles. The outer semicircle is the bridge. The inward semicircle lies in the inner X piece and gives a specified homotopy between the bridge's endpoint maps.

Using that homotopy to identify the endpoints replaces the bridge by $F_0 \times S^1$, attached along its inward section. Its circle is precisely the outer bridge followed by the inward return, hence the actual normal circuit. Finally $F_0 \rightarrow F$ and $F_0 \times S^1 \rightarrow F \times S^1$ are homotopy equivalences. This also accounts for the proper edge collars and proves the assertion about the side maps, not just the absolute homotopy type of E . $\square$

For each j , regard a solid torus V_j as a compression body from the punctured torus F_j to a disk, compressing a_j . Define

$$D = E \cup \coprod_j (F_j \times S^1) \coprod_j (V_j \times S^1).$$

All gluings use the just specified product framings; round corners. The remaining disk face of ∂V_j contributes a solid torus to the new boundary. Hence $S = \partial D$ is the surface-framed surgery of ∂X on the initial grope-boundary link. In particular it is connected.

By Lemma 2.2,

$$D \simeq X \cup \coprod_j F_j \coprod_j (V_j \times S^1).$$

Each $F_j \rightarrow X$ is null-homotopic. The extra summand is therefore the homotopy cofiber of $F_j \rightarrow V_j \times S^1$. Retract the latter space to the torus on its longitude b_j and normal meridian t_j . The cofiber adds one two-cell killing b_j and one on the null loop a_j . Cancel the b_j one/two-cell pair. The torus two-cell then has null attaching map, so the result is

$$S_{t_j}^1 \vee S^2 \vee S^2.$$

Thus $H = \pi_1 D$ is the original free graph group with one new free meridian t_j per surface, and D has the graph-and-spheres homotopy type in Proposition 2.1.

2.3 The middle form and the Poincaré pair

Close f_j by a meridian disk of V_j at its fixed circle coordinate, obtaining a sphere u_j . The two compression framings agree, so its second Stiefel–Whitney evaluation is zero. The g_j remain in E ; therefore (2) gives

$$\lambda(u_i, g_j) = \delta_{ij}, \quad \lambda(g_i, g_j) = 0 \quad \text{over } \mathbb{Z}H.$$

The hermitian matrix of the u_i has even diagonal. As H has no element of order two, every such diagonal is $z + \bar{z}$: choose one representative from each nonidentity inverse pair and halve the even identity coefficient. Add suitable combinations of the g_j to the u_i , taking a triangular half off the diagonal and this half on it. The resulting spheres, denoted a_i , together with $b_i = g_i$, have hyperbolic intersection matrix. The usual framed representative adjustments give zero Wall self-intersections: choose normal Euler number zero using the evenness, and then use $\lambda(a_i, a_i) = \mu(a_i) + \overline{\mu(a_i)}$ and the absence of order-two elements. This is the ordinary immersed-sphere calculus of [9]; no disk-embedding assertion enters it.

These $2k$ classes form a basis. Indeed, $\pi_2 D$ is already known to be free of rank $2k$. Its square matrix of the displayed classes has a left inverse obtained from their nonsingular pairing. To see directly that it also has a right inverse, pass to each finite quotient of H . The resulting matrices act on finite-dimensional complex vector spaces, where a left inverse is a right inverse. Residual finiteness of the free group separates the finite support of any nonzero group-ring element, so the right-inverse identity holds over $\mathbb{Z}H$ itself. We henceforth use this specified basis.

Attach a three-cell on every a_i and b_i to form P . A map from the graph with $2k$ two-spheres to D realizing this basis is a homotopy equivalence: it induces an isomorphism on π_1 and on the homology of simply connected covers. Thus $P \simeq BH$.

We check Poincaré duality for the pair, since this step must not be confused with embedded surgery. Set $\Lambda = \mathbb{Z}H$. In the duality of (D, S) , the middle sphere module maps isomorphically onto a split summand of $H_2(D, S; \Lambda)$, by its nonsingular form. Attaching the three-cells quotients

both absolute and relative homology by these middle summands; the attaching map is injective on them, so no new third homology appears. On cohomology the pullbacks identify the new groups with the annihilators of the sphere summands. If $f : (D, S) \rightarrow (P, S)$ is the natural cell-attachment inclusion and $[P, S] = f_*[D, S]$, cap-product naturality reads

$$z \frown [P, S] = f_*(f^*z \frown [D, S]).$$

Hence the old duality isomorphisms induce exactly the new ones on these annihilators and quotients. The spaces have finite free chain complexes over Λ , so the regular-coefficient cap-product criterion gives Poincaré duality with arbitrary local coefficients. This proves parts (1) and (2) of Proposition 2.1.

Corollary 2.3. *The map $\pi_1 S \rightarrow H$ is onto. If $r = \text{rank } H$, then $H_1(S; \mathbb{Z}) \rightarrow H_1(P; \mathbb{Z})$ is an isomorphism and $H_2(S; \mathbb{Z}) \cong \mathbb{Z}^r$.*

Proof. Duality and the graph type of P give $H_1(P, S; \mathbb{Z}H) = H_0(P, S; \mathbb{Z}H) = 0$. Therefore $H_0(S; \mathbb{Z}H) \rightarrow H_0(P; \mathbb{Z}H)$ is an isomorphism. The former is free abelian on the cosets of the image of $\pi_1 S$, and the latter is $\mathbb{Z}$; there can be only one coset. With ordinary coefficients the pair sequence and duality give $H_2(P, S) = H_1(P, S) = 0$ and $H_3(P, S) \cong H^1(P) \cong \mathbb{Z}^r$. The assertions follow. $\square$

2.4 Topological fillings and the cut obstruction

The construction and duality calculation establish the first two parts of Proposition 2.1. For the third part we must analyze an arbitrary topological filling with the specified boundary map. The following cut construction supplies the hypotheses recalled in Section 2.1; it does not presume a smooth structure on that filling.

Suppose a marked filling D' as in part (3) existed. Attach the dual two-handles that undo the surgery on the initial link. Denote the result by W , so $\partial W = \partial X$. Each attaching circle represents its prescribed new free generator t_j under the marking $D' \simeq BH$. The corresponding one/two-cell pair cancels in the homotopy type. Thus W has the graph type of X , with the original boundary-to-graph map: equality of the marked fundamental-group maps gives equality up to homotopy because the graph is aspherical. The two-handle cocores are disjoint proper locally flat disks bounded by the original initial link. In particular all the original marked words are now represented by disk boundaries.

Here is the cut construction with the hypotheses on W made explicit. Use the marked quotient $\pi_1 W \rightarrow F_m$ and its associated tree cover. Prescribe the equivariant map to be constant at the appropriate vertex on each outside face K_v and each lifted disk neighborhood, and to cross the appropriate edge on each boundary torus collar. These prescriptions agree where they meet. A section of the associated bundle with contractible tree fiber extends them to the rest of W . The free boundary extra loops, with whiskers, generate the fundamental group of this tree cover, just as for the marked graph model of X .

We must allow W to be topological. By Quinn's punctured smoothing theorem [18, Corollary 2.2.3], delete an interior point away from the disks and extend the given boundary smoothing to the complement. Prescribe the map to be constant near this point before extending the section. Approximate it smoothly over the open middle intervals of the finitely many graph edges, away from the disks and puncture, and choose regular values. The cuts are compact smooth three-manifolds, disjoint from the disks. Their filled complementary pieces are topological four-manifolds, smooth away from the finite set of lifted punctures in each finite quotient.

For completeness, connectedness is not automatic from a regular-level construction. Form the component graph of the cuts in the tree cover. Its fundamental group is a quotient of that of the cover, by joining successive crossings through connected complementary pieces. Every extra boundary generator lies in one piece, so maps trivially to this graph. It follows that the component graph is a tree. The outside faces and torus collars give an embedded principal copy of the original plumbing tree: different vertex labels lie in different components of the complement of the target midpoints. Any remaining branch has a unique nearest principal vertex. Its stabilizer is trivial, since it fixes that vertex. The quotient has finitely many components; hence these branches have finite valence and uniformly bounded depth, and each is finite. Ignore their closed cuts and merge them into their principal vertices. This leaves connected vertex pieces and one connected cut with each prescribed torus as its entire boundary. The disk prescriptions have not changed. This is precisely the argument of Lemma 2.5 (Connected tree cuts) of [16], now applied to the marked W rather than requiring a homeomorphism $W = X$.

Pass to the parity quotient, sending every plumbing generator to the nonidentity element of $\mathbb{Z}/2$. Its filled ambient space W_2 has the homotopy type of the graph with two vertices and $2m$ edges, and p extra circles at each vertex. Write A, B for the filled vertex pieces and $Y_1, \dots, Y_{2m}$ for their cuts. Over any characteristic-zero field K , Mayer–Vietoris gives

$$H^2(A; K) \oplus H^2(B; K) \xrightarrow{\cong} \bigoplus_i H^2(Y_i; K), \quad (7)$$

because $H^2(W_2; K) = H^3(W_2; K) = 0$. In degree one the difference of restrictions is onto. Its kernel is the image of $H^1(W_2; K)$ modulo the $2m - 1$ graph classes, and evaluation on the p named extra loops at each vertex identifies that kernel with K^{2p} . Thus the restriction map becomes an isomorphism on the subspace where these extra evaluations vanish.

On a cut Y_i , half-lives/half-dies says that the image of $H_1(\partial Y_i; K)$ in $H_1(Y_i; K)$ is a line. If both original coordinate loops had nonzero image, their images would be nonzero multiples. A class from either vertex annihilates the longitude bounding in its outside face, and hence would annihilate both coordinates. The degree-one surjectivity just proved would then force every class on Y_i to do so, a contradiction. Hence one original coordinate β_i dies and the other α_i survives. The plus endpoint is the side at which β_i is a longitude. The parity involution pairs the two lifts of each join and exchanges their plus endpoints, giving exactly m active edges at A .

The prescribed disks lie in their own vertex pieces. They therefore give every law (1) at those vertices, with all the independently chosen conjugators and signs. The marked ambient free group still has trivial port meridians and freely assignable named extra letters. In particular it supplies the common flat background used in the Vertex data Proposition: assign eight extra letters the eight nonidentity matrices among $\exp(a\sigma e)\exp(b\sigma f)$, $a, b \in \{0, 1, 2\}$, for the standard nilpotent generators e, f of $\mathfrak{sl}_2$. Together with the identity, their adjoint operators span all endomorphisms over $\mathbb{C}((\sigma))$, as in [16, Proposition 6.1]. Thus the geometric and marked hypotheses of the tensor interface in Section 2.1 have all been established.

It remains to justify that its vertex calculation works for the punctured smooth structures just used. We give this step rather than assuming that the filled vertices admit smooth structures.

Lemma 2.4 (Forms on a punctured vertex). *Let V be one of the filled compact vertex pieces and let Z be its finite set of punctures. In degree zero take smooth functions on $V \setminus Z$ separately constant near each puncture, and in positive degrees take smooth forms vanishing near the punctures. This differential graded algebra computes $H^*(V; \mathbb{C})$, naturally on the unpunctured faces. It supports the vertex calculation in the Vertex data Proposition of [16], with the filled cohomology groups and with the same Stokes identities.*

Proof. Regard these forms as a sheaf on V . Its stalk at a puncture is $\mathbb{C}$ in degree zero and zero in higher degrees; elsewhere it is the ordinary de Rham resolution. Partitions of unity can be chosen constant on smaller neighborhoods of the finitely many punctures. The complex is therefore a fine resolution of the constant sheaf, so computes the filled cohomology. A based condition in degree zero removes only H^0 . Linear splittings onto cohomology and contracting homotopies may consequently be chosen inside this complex, exactly as for the ordinary vertex de Rham complex.

Wedges, Lie brackets, the curvature recursions, and differentiation in the formal parameters preserve this algebra. Each fixed positive-degree coefficient vanishes near Z , so integration and Stokes take place on a compact smooth subdomain containing its support. The additional end boundaries contribute zero. An entire formal series need not have one common vanishing neighborhood: every calculation at a fixed coefficient uses only finitely many terms. The source's formal lifts and divisions likewise use finite linear combinations coefficientwise, and extensions from face collars can be supported away from Z .

Removing interior points in dimension four does not change the fundamental group. At each finite nilpotent order the flat systems in question vanish near the punctures in their local trivializations; gauges between them can be separately constant at those ends. Thus the based holonomy and exact-gauge arguments use the same marked group as the filled vertex. Allowing separate constants at distinct ends is important here. These are the uses of vertex smoothness in the proof of Proposition 6.1 of [16]. Its edge-algebraicity theorem, [16, Theorem 5.1], is still applied only to the unchanged compact smooth three-manifolds Y_i . $\square$

Apply Lemma 2.4 to the two vertex pieces. Equation (7) and the degree-one calculations refer to their filled cohomology, which the lemma computes. The Vertex data Proposition, [16, Proposition 6.1], now produces exactly (3)–(6). The Finite tensor obstruction contradicts their existence. Therefore D' cannot exist, completing the proof of Proposition 2.1.

3 Reflection and the duality group

We now close the chamber without losing its labelled middle-dimensional classes, using the reflection framework of Davis [5, Sections 13 and 15, especially Remark 15.9]. A second construction will give the resulting group a proper cocompact action on a CAT(0) space. These are different developments: the first retains the four-dimensional chambers, whereas the second is used only for the assembly theorems.

The input from Proposition 2.1 is a compact connected oriented smooth four-manifold D , with connected boundary S and free fundamental group H , together with a map of pairs

$$(D, S) \longrightarrow (P, S).$$

Here D has the homotopy type of a finite graph with finitely many two-spheres, its $\mathbb{Z}H$ -intersection form has a specified framed hyperbolic basis of k planes, and P is obtained by attaching three-cells along all $2k$ basis spheres. The pair (P, S) is an oriented Poincaré pair of dimension four and $P \simeq BH$. The map on fundamental groups is the given identification with H . We use a collar on S in both spaces, inserting a mapping cylinder on the P side if necessary.

By Corollary 2.3, the boundary map $\pi_1(S) \rightarrow H$ is surjective. In particular its associated cover of S is connected.

3.1 Panels and finite developments

Choose a finite flag triangulation $\mathcal{L}$ of S , for example a barycentric subdivision of a smooth triangulation. Write V for its vertex set and S_s for the closed dual block of $s \in V$ in the barycentric subdivision. These blocks are three-dimensional panels. For nonempty $T \subseteq V$, the intersection $S_T = \bigcap_{s \in T} S_s$ is nonempty exactly when T spans a simplex of $\mathcal{L}$; in that case it is a ball of dimension $4 - |T|$. Collar neighborhoods of these intersections have the usual orthant structure.

Let

$$C = \langle s \in V \mid s^2 = 1, \quad st = ts \text{ whenever } \{s, t\} \in \mathcal{L} \rangle$$

be the right-angled Coxeter group. For a collared space B with these panels in its boundary, its development is

$$\begin{aligned} \mathcal{U}(C, B) &= (C \times B) / \sim, & T(x) &= \{s : x \in S_s\}, \\ (c, x) &\sim (c', x) & \iff & c^{-1}c' \in C_{T(x)}. \end{aligned}$$

Here C_T is the subgroup generated by T and $T(x) = \emptyset$ off the boundary. Each chamber embeds: the equivalence relation never identifies different values of x . For nonempty $T(x)$ the subgroup $C_{T(x)}$ is $(\mathbb{Z}/2)^{|T(x)|}$, and its sectors fill the corresponding orthant neighborhood.

We use the elementary Coxeter word facts that special subgroups embed, a finite special coset has a unique shortest element, and the right descent set

$$\text{Des}(c) = \{s \in V : |cs| < |c|\}$$

is a clique. These follow from the right-angled reduced-word rule of commuting adjacent commuting letters and cancelling equal consecutive letters; see also [4, Chapters 4–5].

Lemma 3.1 (Labelled finite developments). *Put $Y = \mathcal{U}(C, D)$ and let D_c be its chamber with label c . Order C by nondecreasing word length, breaking ties arbitrarily. Every finite initial segment containing the identity is, after rounding its collar corners, an iterated boundary connected sum of its chambers. In particular*

$$U_L = \bigcup_{|c| \leq L} D_c$$

is a compact manifold with boundary the connected sum of the corresponding copies of S . Its fundamental group is the free product of their chamber groups. The same statements hold if any chambers have been stabilized in their interiors.

Proof. The part of a new chamber D_c already present is exactly

$$A_c = \bigcup_{s \in \text{Des}(c)} S_s. \tag{8}$$

To verify the possible equal-length overlaps, consider a point with panel set T . Its incident chambers are the special coset cC_T . Write a for the shortest element of that coset. Its members have the form $a \prod_{s \in J} s$, with $J \subseteq T$, and length $|a| + |J|$. If a distinct incident chamber occurs no later than c , then c is not the shortest member, so it has a descent in T . This places the point in (8). The converse follows from the shorter adjacent chamber across each descent panel.

The set in (8) is a three-ball. More generally a union of dual panels indexed by a nonempty simplex is a ball. One proves this by adding its panels one at a time. A new panel is a ball, and its intersection with the preceding panels is the corresponding union of dual panels in its boundary

link. Induction on dimension makes that intersection a ball of one lower dimension. The dual-block coordinates give the product collars needed to glue the two balls. This argument also applies to panels in any simplicial cover of S .

It remains to check that A_c is a ball in the *boundary* of the preceding union. Contractibility alone would not establish the claim. At a face incident to q panels, label the 2^q local orthants by subsets of those panels, starting at the shortest incident chamber. The already present orthants form a Boolean downset I : every proper subset of a present subset has smaller word length. This remains true with arbitrary choices within a length tie.

When I is nonempty and proper, its frontier is a PL hypersurface. For an explicit verification, write local coordinates as $x = u + t(1, \dots, 1)$ with $\sum u_i = 0$, and label an orthant by its positive coordinates. Its union is described by

$$t \leq f(u), \quad f(u) = \max_{A \text{ maximal in } I} \min_{i \notin A} (-u_i).$$

The function f is finite, continuous, and piecewise linear. Hence the frontier is locally a graph. Along the incoming faces some sectors are present and the new chamber sector is absent, so these faces lie on this frontier. The panel identifications are PL, and the embedded incoming three-ball has locally flat boundary in this three-manifold frontier. Gluing the new chamber along it is therefore a boundary connected sum, after rounding. Induction proves the manifold and boundary assertions. Van Kampen gives the labelled free product, and interior stabilizations do not affect any collar argument. $\square$

3.2 The universal development

Let

$$\Delta = \ker(C \longrightarrow (\mathbb{Z}/2)^V),$$

where each generator maps to its own coordinate vector, and define

$$N = \Delta \backslash \mathcal{U}(C, D), \quad Q = \Delta \backslash \mathcal{U}(C, P).$$

The parity map is injective on every finite special subgroup. Thus Δ has trivial intersection with every point stabilizer in either development. The quotient has finitely many chambers. For D the orthant charts show that N is a closed four-manifold. Give chamber c the orientation $(-1)^{|c|}$ times the chosen chamber orientation. Adjacent chamber orientations fit across a panel, and Δ preserves them. Thus N is oriented. The same convention orients the reflected Poincaré pairs.

Let $\widehat{S} \rightarrow S$ be the connected cover determined by $\pi_1(S) \rightarrow H$, with lifted triangulation $\widehat{\mathcal{L}}$. This is a flag triangulation. Indeed, the vertices of a lifted clique project to distinct pairwise adjacent vertices downstairs. They span a simplex there; lifts of its edges, and of its triangular relations, place all the vertices in one lift of that simplex. Denote by $\widehat{C}$ the right-angled Coxeter group on $\widehat{\mathcal{L}}$. The deck action of H permutes its generators. Forgetting the lift gives an H -invariant homomorphism $q : \widehat{C} \rightarrow C$, and hence a surjection

$$\begin{aligned} \rho : \widehat{C} \rtimes H &\longrightarrow C, & \rho(\widehat{c}, h) &= q(\widehat{c}), \\ G &= \rho^{-1}(\Delta). \end{aligned} \tag{9}$$

Let $\widetilde{P}$ be the universal cover of P , with boundary preimage $\widehat{S}$ and its lifted panels. On $\widehat{Q} = \mathcal{U}(\widehat{C}, \widetilde{P})$ the action is

$$(\widehat{c}, h)[\widehat{d}, x] = [\widehat{c}h(\widehat{d}), hx].$$

The map $[\widehat{d}, x] \mapsto [q(\widehat{d}), \bar{x}]$ descends to the ordinary development. On a chamber interior it is the usual covering map. At a boundary point it is also a covering map: the lifted panel set maps bijectively to its downstairs panel set, and the corresponding finite special groups map isomorphically. These local charts identify $\widehat{Q}/G$ with Q . The action of G is free: if an element fixes a point, its H -component fixes the chamber coordinate and is therefore trivial, after conjugating the chamber label; the remaining stabilizer is a lifted finite special group, on which ρ is injective and whose image misses Δ .

The space $\widehat{Q}$ is contractible. Filter its chambers by word length in $\widehat{C}$. The first chamber $\tilde{P}$ is contractible. Every further chamber meets the shorter ones in the nonempty ball of its descent panels, by the argument of Lemma 3.1. Its attachment is therefore a homotopy equivalence. In an infinite word-length layer, new interiors are disjoint and their mutual intersections already lie in shorter layers. All these attachments may be performed simultaneously. The CW direct limit is weakly contractible and hence contractible. This proves, in particular, that Q is a finite $K(G, 1)$.

Proposition 3.2 (The reflected pair). *The space Q is a finite oriented Poincaré complex of dimension four. The natural map $f : N \rightarrow Q$ has degree one and induces an isomorphism on fundamental groups, identifying both with the group G in (9). If b is the total number of chamber hyperbolic planes in N , then*

$$\pi_2(N) \cong (\mathbb{Z}G)^{2b}$$

has the specified orthogonal hyperbolic basis, and $\tilde{H}_i(\tilde{N}; \mathbb{Z}) = 0$ for $i \neq 2$. The group G is finitely presented, is torsion-free, and satisfies

$$H^i(G; \mathbb{Z}G) = 0 \quad (i \neq 4), \quad H^4(G; \mathbb{Z}G) \cong \mathbb{Z} \quad (10)$$

with trivial right G -action.

Proof. Use the same universal development with $\tilde{D}$ in place of $\tilde{P}$. The preceding covering argument is unchanged. The first chamber is simply connected; attaching subsequent simply connected chambers along balls shows that this development is the universal cover $\tilde{N}$. Its reduced homology is the direct sum of the chamber reduced homologies, each concentrated in degree two. There are finitely many chamber orbits in N . For each one, induction from its chamber subgroup gives

$$\mathbb{Z}G \otimes_{\mathbb{Z}H} \pi_2(D) \cong (\mathbb{Z}G)^{2k}.$$

The spheres lie in chamber interiors, so different chambers are orthogonal. Within a chamber the form is its given $\mathbb{Z}H$ -form, extended to $\mathbb{Z}G$. Reversing one vector in a plane absorbs any chamber orientation sign. This gives the asserted basis and form.

The construction of P attaches exactly the indicated three-cells away from the boundary. Consequently Q is obtained from N by attaching three-cells on this entire middle basis. It follows at once that f induces the displayed identification of fundamental groups. These attachments split off a nonsingular middle summand in both absolute and relative duality: on homology they quotient by the sphere summand and on cohomology they restrict to its annihilator. With $[Q] = f_*[N]$, naturality of cap product identifies the surviving duality maps with those of Q . The finite free chain criterion therefore gives Poincaré duality for Q , just as for the chamber pair. Equivalently one may glue the chamber duality maps across the manifold collar sectors, where the orientations cancel in pairs. The top fundamental class is unchanged, so f has degree one.

For clarity, the regular-coefficient calculation is particularly short. In the cohomology sequence for these three-cell attachments, the only potentially new map is evaluation

$$H^2(N; \mathbb{Z}G) \longrightarrow \text{Hom}_{\mathbb{Z}G}(\pi_2(N), \mathbb{Z}G).$$

Under duality it is the adjoint of the nonsingular hyperbolic form, and hence an isomorphism. The middle cohomology thus disappears, whereas $H^4(Q; \mathbb{Z}G) = H^4(N; \mathbb{Z}G) = \mathbb{Z}$; all other degrees vanish. The right action on this last group is trivial because the orientation chosen above is preserved by G .

Finally, finiteness and asphericity of Q give a finite presentation and a finite free resolution of $\mathbb{Z}$ from the chains of $\widehat{Q}$. They also give finite integral cohomological dimension, which excludes torsion: restriction to a nontrivial finite cyclic subgroup would give a finite projective resolution for that group, contrary to its nonvanishing cohomology in arbitrarily high degrees. This proves all the assertions, including (10). $\square$

3.3 A proper cocompact CAT(0) model

The lifted nerve has infinitely many vertices when H is infinite. Its Davis complex alone need not give a cocompact action of the semidirect product. The following support construction addresses this.

Lemma 3.3. *The group $\widehat{C} \rtimes H$, and hence its finite-index subgroup G , acts properly and cocompactly by isometries on a proper CAT(0) cube complex of dimension at most five.*

Proof. Choose a locally finite tree T on which the free group H acts freely and cocompactly, without inversions. For the trivial group one may take a single vertex. Assign to each lifted-nerve vertex s a point $a_s \in T^{(0)}$, equivariantly under H . There are finitely many vertex and edge orbits in the lifted nerve. Thus the distances $d_T(a_s, a_t)$ for adjacent s, t have a common finite bound. Choose an integer r large enough that the closed radius- r subtrees T_s about a_s intersect for every such pair.

For a vertex v and an edge e of T , put

$$A_v = \{s : v \in T_s\}, \quad A_e = \{s : e \subset T_s\}.$$

These sets have uniformly bounded finite size: there are finitely many generator orbits, and a fixed-radius ball in the free cocompact tree contains uniformly finitely many orbit points. Let W_v and W_e be the special subgroups of $\widehat{C}$ generated by A_v and A_e . For $e = [v, w]$ we have $A_e = A_v \cap A_w$, so the edge maps are special subgroup inclusions.

The graph of groups over the tree T has colimit $\widehat{C}$. Indeed, occurrences of a generator s are identified precisely over its connected support T_s . Every defining commutation of $\widehat{C}$ occurs in some vertex group, because the two supports intersect at a vertex. Conversely, each vertex group uses the *induced* subgraph of the lifted nerve, and so introduces no additional relation. The resulting presentation is exactly the right-angled presentation of $\widehat{C}$. All the edge maps are injective, as are the maps of these special subgroups into $\widehat{C}$.

Let Σ_v and Σ_e be the Davis cube complexes of these finite-generator right-angled groups. They are CAT(0), and the special subcomplexes $\Sigma_e \subset \Sigma_v$ are convex in the standard piecewise Euclidean cube metric. The CAT(0) assertion is [4, Theorem 12.2.1(i)]; special-subgroup metric convexity is [20, Appendix, Proposition A.2]. Form a tree of spaces with vertex spaces $\widehat{C} \times_{W_v} \Sigma_v$ and edge cylinders $\widehat{C} \times_{W_e} (\Sigma_e \times [0, 1])$. Its underlying tree is the Bass–Serre tree, whose vertices and edges are the cosets $\widehat{C}W_v$ and $\widehat{C}W_e$, with their indicated labels in T ; it is not simply the original support tree. Glue each cylinder at its endpoints by the special-subcomplex maps. Convex gluing gives a CAT(0) space over every finite subtree, preserving the factors as convex isometric subspaces [2, Theorem II.11.1 and Remark II.11.2]. One can apply this gluing first over finite subtrees and then pass to the union; the local finiteness verified next ensures completeness.

There is no local-finiteness problem from potentially infinite Bass–Serre valence. In a fixed vertex Davis complex, for any one incident support edge, translates of its special subcomplex have vertex sets equal to the cosets of W_e in W_v . These cosets partition the Cayley vertices. A finite subcomplex therefore meets only finitely many of these translates. Each support vertex has only finitely many incident edges. Since each Σ_v is locally finite, the whole tree of cube complexes is locally finite. Its cubes have unit edge lengths and uniformly bounded dimension, so the induced length metric is proper and complete.

Left multiplication gives the $\widehat{C}$ -action. The equivariance of the supports extends this to $\widehat{C} \rtimes H$. An element stabilizing a vertex space projects to an element of H fixing its support vertex, and that element is trivial. Within the vertex space its point stabilizer is therefore a finite Coxeter stabilizer. The same reasoning applies to edge interiors. The cellular action on this locally finite complex is proper. There are finitely many H -orbits of support vertices and edges, and each corresponding Davis complex has compact quotient by its special group. Hence the full action is cocompact. Each clique of the lifted nerve has at most four vertices; the vertex spaces have dimension at most four and the edge cylinders dimension at most five. Finally G has finite index by (9), so its restricted action has all the same properties. $\square$

It follows that G satisfies the K - and L -theoretic Farrell–Jones conjectures with coefficients: use Bartels–Lück [1, Theorem B(ii)] and Wegner [22, Theorems 1.1 and 3.4]. In particular, because G is torsion-free, the Whitehead group, reduced $K_0(\mathbb{Z}G)$, and negative K -groups vanish, and the simple L -theory assembly map is an isomorphism in every degree; see [1, Proposition 0.3(i)]. These are the assembly consequences used below. No four-dimensional rigidity or disk-embedding conclusion is being inferred from CAT(0) geometry.

3.4 The gallery map

We finish by recording the geometric map needed in the truncation argument. The preceding CAT(0) model supplies assembly; the map here instead records positions and directions in the ordinary chamber development. Let Σ_C denote the ordinary Davis cube complex of C . Its vertices are the chamber labels C , and the link of the identity vertex is $\mathcal{L}$, whose realization is S .

Lemma 3.4. *There is a C -equivariant continuous map*

$$\eta : Y \longrightarrow \Sigma_C$$

which sends the inner core of each chamber to its label, respects the panels, and has uniformly bounded image diameter on each chamber. On the boundary collar of the central chamber its radial directions at the identity vertex give the dual-block identification with S . After interior chamber stabilizations, the same construction gives a continuous map with the same panel, core, and diameter properties; it is equivariant under any subgroup preserving the stabilized pattern.

Proof. The Davis chamber is the cone on the barycentric subdivision of $\mathcal{L}$, with its usual dual panels on the base. Identify that base panelwise with S . Map a boundary collar of D to this cone by that identification on its outer boundary and by coning to the apex on its inner boundary; map the remainder of D to the apex. The maps agree across reflected panels and hence define η . Every chamber image lies in the corresponding compact Davis chamber, whose diameter is independent of its label. At the identity apex, radial projection of the cone base to a sufficiently small metric sphere gives exactly its link identification. Stabilization changes only the collapsed inner core, so does not affect the construction. $\square$

4 Marked stable realization

We now suppose, for a contradiction, that the group constructed above is the fundamental group of a closed aspherical topological four-manifold M . The preceding construction gives a finite oriented Poincaré four-complex $Q = BG$ and a closed oriented manifold N . Write q for the map of Proposition 3.2:

$$q: N \longrightarrow Q$$

which is an isomorphism on fundamental groups and has degree one. The only reduced homology of the universal cover of N is its second homology. With $\Lambda = \mathbb{Z}G$, this is the free module

$$A = \pi_2(N) \cong \Lambda^{2b}, \quad \lambda_N \cong \bigoplus_{a=1}^b \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (11)$$

The displayed basis consists of the prescribed chamber sphere pairs. All identifications include whiskers and orientations. Moreover, G is torsion-free, is an integral PD_4 group, and satisfies the K - and L -theoretic Farrell–Jones conjectures by the $\text{CAT}(0)$ construction. These are conclusions of the construction, not additional assumptions on a putative realization.

Both Q and M are $K(G, 1)$ spaces. Their equivalence identifies the orientation character with the action on the top dualizing module. Consequently M is orientable. Choose its orientation and its fundamental-group marking so that its classifying map $c_M: M \rightarrow Q$ has degree one, and put

$$L = M \# b(S^2 \times S^2).$$

Collapsing the simply connected summands gives a degree-one classifying map $c_L: L \rightarrow Q$. Identify $\pi_2(L)$ with A by sending each specified chamber pair in (11) to a specified standard pair in L . This is an isometry of equivariant forms. We must retain this particular isometry throughout the argument.

4.1 An integral marked homotopy equivalence

We first realize the prescribed module isometry by a homotopy equivalence. The comparison takes place in a common second Postnikov stage: group duality splits that stage, and integral cup-product tests identify the two fundamental classes there. These are the inputs to the lifting argument below.

Lemma 4.1. *The second Postnikov stages of N and L , with the preceding markings, identify over Q with the split stage*

$$B = EG \times_G K(A, 2).$$

There are maps $u_N: N \rightarrow B$ and $u_L: L \rightarrow B$ inducing the specified identifications of fundamental and second homotopy groups. If $\Gamma(A) = H_4(K(A, 2); \mathbb{Z})$, then

$$0 \longrightarrow \Gamma(A)_G \longrightarrow H_4(B; \mathbb{Z}) \longrightarrow H_4(Q; \mathbb{Z}) \longrightarrow 0 \quad (12)$$

is a split exact sequence.

Proof. Since A is finite free over Λ , integral group duality gives

$$H^2(G; A) = H^3(G; A) = 0. \quad (13)$$

The second Postnikov invariant lies in the latter group, so it vanishes. The former removes the degree-two ambiguity in making the identification with the split stage for the prescribed module marking. We use a functorial Eilenberg–Mac Lane model for $K(A, 2)$, so the given action on A defines the displayed homotopy quotient and its section.

Separately, freeness gives $H_i(G; A) = 0$ for $i > 0$. In total degree four, the homology spectral sequence of

$$K(A, 2) \longrightarrow B \longrightarrow Q$$

therefore has only $H_4(G; \mathbb{Z})$ and $H_0(G; \Gamma(A))$ as possibly nonzero terms: the intervening term $H_2(G; A)$ is zero, and the fiber has no homology in degrees one and three. The possible incoming differentials to $H_0(G; \Gamma(A))$ have sources $H_3(G; A) = 0$ and $H_5(G; \mathbb{Z}) = 0$. The section preserves the base term. This proves (12), including its integral, rather than merely rational, exactness. $\square$

The next elementary check is useful because the middle classes are specified over the group ring. Ordinary intersection forms would lose the relative translates needed here.

Lemma 4.2. *With the identifications above,*

$$(u_N)_*[N] = (u_L)_*[L] \quad \text{in } H_4(B; \mathbb{Z}). \quad (14)$$

Proof. Write $e_1, \dots, e_{2b}$ for the chosen Λ -basis of A and $\phi_i: A \rightarrow \Lambda$ for the coordinate functionals. The low-degree cohomology sequence of the Postnikov fibration, with regular coefficients, contains

$$0 \longrightarrow H^2(G; \Lambda) \longrightarrow H^2(B; \Lambda) \longrightarrow \text{Hom}_\Lambda(A, \Lambda) \longrightarrow H^3(G; \Lambda).$$

Both outside groups vanish. Thus each ϕ_i is the restriction of a unique class $x_i \in H^2(B; \Lambda)$.

Their cup products may be evaluated without multiplying away either coefficient factor:

$$z \longmapsto \langle x_i \smile x_j, z \rangle \in (\Lambda \otimes_\mathbb{Z} \Lambda)_G, \quad z \in H_4(B; \mathbb{Z}). \quad (15)$$

The tensor product has the diagonal left action. Its coinvariants identify, as an abelian group, with Λ by $g \otimes h \mapsto g^{-1}h$. For a closed oriented manifold with the indicated middle module, evaluations in (15) are the equivariant cup form on the coordinate functionals. By Poincaré duality this is the dual form of its equivariant intersection pairing. The prescribed isometry therefore makes these evaluations equal on $(u_N)_*[N]$ and $(u_L)_*[L]$.

For completeness, these tests detect the fiber term integrally. The underlying abelian group of A is free on the elements ge_i . The group $\Gamma(A)$ has a basis consisting of one square generator for each such element and one cross generator for each unordered pair of distinct such elements. The action of G permutes this basis. Hence its coinvariants are free abelian on the corresponding orbits. A nonidentity element stabilizing an unordered pair would exchange its two members and have order two; torsion-freeness rules this out.

The square orbit represented by e_i evaluates to 1 in the (i, i) test. For $i < j$, the cross orbit represented by $\{e_i, te_j\}$ evaluates to t in the (i, j) test. Finally, for $t \neq 1$, the orbit represented by $\{e_i, te_i\}$ evaluates to $t + t^{-1}$ in the (i, i) test. These last orbits are indexed by t modulo inversion. Since G has no elements of order two, their supports are disjoint two-element subsets of $G \setminus \{1\}$. The displayed tests are therefore jointly injective on $\Gamma(A)_G$. In particular, no integral square or hidden two-torsion is discarded.

The two fundamental classes have the same image $[Q]$ in $H_4(Q; \mathbb{Z})$. By Lemma 4.1, their difference lies in $\Gamma(A)_G$. All its cup tests vanish, so the difference is zero. $\square$

Proposition 4.3. *There is an orientation-preserving homotopy equivalence*

$$f: N \longrightarrow L$$

inducing the specified isomorphism on π_1 and the prescribed isometry $A \rightarrow \pi_2(L)$.

Proof. Replace $u_L: L \rightarrow B$ by a fibration. Its fiber is two-connected, with third homotopy group $\pi_3(L)$ carrying the usual G -action. The first obstruction to lifting u_N is consequently the pullback of a universal class

$$\kappa \in H^4(B; \pi_3(L)).$$

The pullback $u_L^* \kappa$ is zero, since u_L lifts to its own fibration replacement. Naturality of cap product and Lemma 4.2 give

$$\begin{aligned} (u_N)_*((u_N^* \kappa) \cap [N]) &= \kappa \cap (u_N)_*[N] \\ &= \kappa \cap (u_L)_*[L] = 0 \quad \text{in } H_0(B; \pi_3(L)). \end{aligned}$$

The map u_N induces an isomorphism on fundamental groups, and hence on these degree-zero coefficient groups. Poincaré duality identifies

$$H^4(N; \pi_3(L)) \xrightarrow{\cap [N]} H_0(N; \pi_3(L)) \cong \pi_3(L)_G.$$

Thus the lifting obstruction vanishes.

The manifold N has a finite four-dimensional CW structure from the smooth chamber construction and the panel collars. There are no higher obstructions on this source. No triangulation or smooth structure on the hypothetical M is required. The resulting lift f has the stipulated maps on π_1 and π_2 . The universal covers of N and L have reduced homology only in degree two, where the lift is an isomorphism. These manifolds have CW homotopy type [15, Corollary 1]. The homological Whitehead theorem on their simply connected universal covers, followed by Whitehead's theorem, shows that f is a homotopy equivalence [10, Corollary 4.33 and Theorem 4.5]. Finally $c_L f \simeq q$ and both classifying maps have degree one, so $f_*[N] = [L]$. $\square$

4.2 From the marked equivalence to a marked stable product

The marked homotopy equivalence is now available. The next two steps pass through an s -cobordism and then a stabilized product, retaining the same identification of every original sphere class. The product trace also specifies the new classes introduced by stabilization.

Lemma 4.4. *There is an s -cobordism $(W; N, L)$ whose two end inclusions identify π_1 and π_2 by the marking of Proposition 4.3.*

Proof. The universal intersection forms are even, so N and L are almost spin. Indeed, evaluation of w_2 on every spherical class is its self-intersection modulo two, and on the simply connected cover these evaluations detect w_2 .

We apply the proof of Kasprowski–Land's Theorem 1.1, specifically Lemma 2.3, Lemma 3.3 and Theorem 3.1 of [11]. All hypotheses hold: Q is an aspherical Poincaré four-complex; G satisfies both Farrell–Jones conjectures; the two manifolds have the same orientation character and degree-one classifying maps; and they are almost spin. Their normal-invariant calculation, using assembly injectivity and the Atiyah–Hirzebruch spectral sequence, identifies the normal invariant of this particular f with a class in

$$\ker(H_2(L; \mathbb{Z}/2) \longrightarrow H_2(Q; \mathbb{Z}/2)).$$

The Serre spectral sequence for the universal-cover fibration shows that this kernel is represented by spherical classes. Since L is almost spin, their w_2 evaluations vanish. Lemma 3.3 of [11] therefore supplies a target self-equivalence ϕ with the same normal invariant as f , of the form

$$L \longrightarrow L \vee S^4 \xrightarrow{\text{id} \vee \eta^2} L \vee S^2 \xrightarrow{\text{id} + a} L.$$

Here $a : S^2 \rightarrow L$ represents the spherical class. This pinch acts identically on π_1 and π_2 , because its extra part factors through the four-sphere. Thus $\phi^{-1} \circ f$ has trivial normal invariant and the same marking as f . Surgery of its normal bordism over L , as in Theorem 3.1 of [11], gives the required s -cobordism. Neither this conclusion nor the cited theorem requires G to be a good group. We have not asserted that W itself is a product. $\square$

We use the stable product theorem for this particular bordism. Quinn's Theorem 1.1 in [19], in the form stated in [12, Remark 3.7], says that a five-dimensional topological s -cobordism becomes a product relative to one end after finitely many connected sums along arcs with $(S^2 \times S^2) \times [0, 1]$. There is no restriction on its fundamental group. The relative product statement is important: an arbitrary stable homeomorphism of the ends would not, by itself, retain the specified sphere classes.

Proposition 4.5. *For some integer $k \geq 0$ there is a homeomorphism*

$$h : N \# k(S^2 \times S^2) \longrightarrow L \# k(S^2 \times S^2)$$

which induces the prescribed original marking and matches every added standard pair with its specified added standard pair. In particular, on the left, all the prescribed original hyperbolic pairs and all new standard pairs have simultaneous geometric representatives with pairwise disjoint punctured $S^2 \times S^2$ neighborhoods.

Proof. Apply the stable product theorem to W from Lemma 4.4. Denote its stabilized ends by N^* and L^* and the stabilized bordism by W^* . We verify its marking, including the new summands.

The operation removes disjoint normal $D^4 \times [0, 1]$ neighborhoods of arcs between the ends and glues

$$((S^2 \times S^2) \setminus \text{int } D^4) \times [0, 1]$$

along the resulting $S^3 \times [0, 1]$ interfaces. Removing these codimension-four arcs changes neither π_1 nor second homology of the universal cover. Mayer–Vietoris across the lifted interfaces therefore gives a specified decomposition

$$\pi_2(W^*) \cong \pi_2(W) \oplus \Lambda^{2k}. \tag{16}$$

The second summand is represented by the standard spheres in the added product pieces. Choose their whiskers along the stabilization arcs. The two end inclusions $j_N : N^* \rightarrow W^*$ and $j_L : L^* \rightarrow W^*$ carry each new pair to this same pair, while their old summands map to $\pi_2(W)$ with exactly the original bordism identification.

A product structure $F : N^* \times [0, 1] \rightarrow W^*$ which is the identity on the incoming end restricts at the other end to a homeomorphism $h : N^* \rightarrow L^*$. Use the product trace for the outgoing basepoint path. Any discrepancy from the original bordism path can be corrected by pushing the outgoing basepoint around a loop in L^* , since j_L is an isomorphism on fundamental groups. The resulting based homotopy given by F implies

$$(j_L)_* h_* = (j_N)_*.$$

Both inclusions are isomorphisms on π_1 and π_2 . Thus $h_* = (j_L)_*^{-1}(j_N)_*$, with the same base paths. In the decomposition (16) it preserves the original marking and identifies the new standard basis element by element. There is no arbitrary automorphism mixing the old and new summands.

All $b + k$ standard pairs on $L^* = M \# (b + k)(S^2 \times S^2)$ have disjoint punctured $S^2 \times S^2$ neighborhoods. Give the two spheres of each pair the common whisker to their intersection used for its module marking, and pull the whole based neighborhood back by h . A pair has its one prescribed transverse intersection; different pairs and their neighborhoods are disjoint. The marking just proved identifies their based homotopy classes with the original chamber basis and the added standard basis, respectively. $\square$

4.3 Periodically lifted geometric data

Let C be the right-angled Coxeter group used in the reflection construction, and let $\Delta \leq C$ be the finite-index torsion-free normal subgroup used to obtain N from its ordinary reflection development. Place the finitely many stabilization balls on the N -side inside chamber interiors. This may be done by moving their endpoints along paths before performing the arc connected sums. Track the new whiskers along those paths. There is no requirement that different chambers receive equal numbers of stabilizations.

Lift the resulting N^* to the ordinary reflection development and call the lifted manifold Y . Its chambers, indexed by $v \in C$, are stabilized copies D_v of D . The numbers of stabilizations are Δ -periodic. Gallery distance is the word metric on the chamber indices for the standard Coxeter generators.

Proposition 4.6. *The manifold Y has two Δ -periodic systems of sphere pairs, with the same chamber labels:*

1. *framed immersed hyperbolic pairs in the interiors of the chambers D_v , including the added standard pairs, with their equivariant products already correct over the individual chamber group;*
2. *geometrically standard pairs with pairwise disjoint punctured $S^2 \times S^2$ neighborhoods, based-homotopic sphere by sphere to the correspondingly labelled pairs in the first system.*

There is an integer d_0 such that the drawing of each pair, its standard neighborhood, and the homotopies relating the two systems are contained in the chambers at gallery distance at most d_0 from that pair's label. Basings in this statement are local basings transported with the chamber; no bound is asserted for whiskers drawn from one global base point to all chambers.

Proof. The original immersed representatives stay in their chamber interiors, and the stabilization spheres can be put in their interior connected summands. Stabilization adds orthogonal standard planes and changes neither the chamber fundamental group nor the old pairings.

Proposition 4.5 supplies a finite collection of geometric pairs on the compact quotient N^* , representing the specified based classes. Choose a based homotopy for each sphere between its two representatives. Lift the neighborhoods, sphere drawings, and homotopies to Y , using the prescribed marking to choose the lift for each chamber label. Covering maps preserve π_2 and lift these based homotopies, so they end at precisely the labelled geometric representatives. Disjoint embedded neighborhoods lift to disjoint embedded neighborhoods.

There are only finitely many labelled data on the quotient. Each chosen lift of a drawing, closed standard neighborhood, local basing, or homotopy has compact image. Local finiteness of the development implies that it meets only finitely many chambers. Take d_0 to be the maximum gallery

distance from its assigned label over this finite list of chosen lifts. All remaining data are translates by Δ , whose action preserves gallery distance. The same d_0 therefore works everywhere. $\square$

The last proposition is the stable information needed below. It gives actual geometric summands with their original chamber labels and compactly supported homotopies to the chamber systems. It makes no claim of controlled isotopy, and no claim that the unstabilized chamber obstruction has disappeared.

5 Controlled removal of the middle homology

Suppose, towards a contradiction, that the group constructed above is the fundamental group of a closed aspherical topological four-manifold. We use the marked systems supplied by Proposition 4.6. Thus, after finitely many stabilizations on the compact quotient, the ordinary reflection development is a manifold

$$Y = \bigcup_{v \in C} D_v.$$

Here D_v is a stabilized copy of the chamber, its boundary is S , and its group is the free group H . Boundary identifications are understood with the signs induced by the orientation of Y ; signs in hyperbolic planes are absorbed by changing one basis vector. There are two labelled systems in Y : the chamber-wise immersed hyperbolic bases, denoted $\mathcal{A}$, and disjoint standard hyperbolic pairs, denoted $\mathcal{B}$. Each pair of $\mathcal{B}$ has an embedded neighborhood homeomorphic to $(S^2 \times S^2) \setminus \text{int } D^4$. Corresponding spheres in the two systems are based-homotopic. All these data, including the homotopies, are lifts of finitely many compact data on the quotient.

We will remove an expanding finite part of $\mathcal{B}$ and use the remaining part of $\mathcal{A}$ to perform surgery. The latter operation requires geometric duals. We obtain them by applying the proof of the disc embedding theorem to a specially constructed null-loop input; we do not assume that H is a good group.

There are three tasks. First we cap the noncentral boundary factors and remove the inner standard pairs, leaving a compact manifold with the original boundary marking. Next we prepare the remaining sphere systems and place their upper grope branches over contractible direction patches. This supplies the null-loop input. Finally the tower construction and sphere surgeries remove the remaining middle homology and produce a graph-type filling.

5.1 Complementary fillings and the peeling operation

Write $r = \text{rank } H$. A stabilization of D has homotopy type

$$\bigvee^r S^1 \vee \bigvee^{2k} S^2$$

for some k , and its middle module has the specified hyperbolic basis. Neither the boundary nor its map to BH changes under stabilization.

Lemma 5.1 (Complementary fillings). *For every such stabilized chamber there is a simply connected compact oriented topological four-manifold J with boundary $-S$ such that*

$$H_2(S; \mathbb{Z}) \longrightarrow H_2(J; \mathbb{Z}) \quad \text{is an isomorphism,} \quad H_3(J; \mathbb{Z}) = 0.$$

Moreover, $D \cup_S J$ is simply connected, and inclusion of D identifies its second homology and intersection form with those of $D \cup_S J$.

Proof. First recall the consequences of the graph-type Poincaré pair (P, S) recorded in Corollary 2.3. Duality and its ordinary homology sequence give

$$H_1(S; \mathbb{Z}) \cong H_1(P; \mathbb{Z}) \cong \mathbb{Z}^r, \quad H_2(S; \mathbb{Z}) \cong H^1(P; \mathbb{Z}) \cong \mathbb{Z}^r. \quad (17)$$

The same corollary gives the surjectivity of $\pi_1(S) \rightarrow H$.

Start with the oppositely oriented copy of D . Represent a free basis of its fundamental group by disjoint embedded circles. Their normal bundles are trivial, and general position makes the circles disjoint from the finite sphere system. Interior surgery on these circles kills the free generators. The old sphere classes retain their hyperbolic intersection and framing data after augmentation to $\mathbb{Z}$. In the now simply connected manifold, the sphere embedding theorem with geometric duals realizes a Lagrangian by disjoint framed embedded spheres. Surgery on these spheres preserves simple connectivity and removes the old hyperbolic planes. This use of the disc theorem is in the simply connected case only; see [17, Theorem B].

Let J be the result. Each circle surgery increases Euler characteristic by two, and each sphere surgery decreases it by two. Hence

$$\chi(J) = (1 - r + 2k) + 2r - 2k = 1 + r.$$

The boundary is connected and J is simply connected. Thus $H^1(J; \mathbb{Z}) = 0$ by the cohomology sequence of the pair, and Poincaré–Lefschetz duality gives $H_3(J; \mathbb{Z}) = 0$. Since J has nonempty boundary, $H_4(J; \mathbb{Z}) = 0$, and therefore $b_2(J) = r$.

There is no integral index ambiguity in this calculation. The relevant part of the duality sequence is

$$0 \longrightarrow H_2(S; \mathbb{Z}) \longrightarrow H_2(J; \mathbb{Z}) \longrightarrow H^2(J; \mathbb{Z}) \longrightarrow H_1(S; \mathbb{Z}) \longrightarrow 0. \quad (18)$$

Here $H_3(J; \mathbb{Z}) = H^1(J; \mathbb{Z}) = 0$ accounts for the initial zero. The universal coefficient theorem makes $H^2(J; \mathbb{Z})$ free, because $H_1(J; \mathbb{Z}) = 0$. Every torsion element of $H_2(J; \mathbb{Z})$ would consequently belong to the image of the injected free group $H_2(S; \mathbb{Z})$, so there is no such element. The first two nonzero groups in (18) now have the same rank r . The middle map has rank zero and takes values in a free group, hence is zero. Exactness proves that the first map is an integral isomorphism.

The map $H_2(S; \mathbb{Z}) \rightarrow H_2(D; \mathbb{Z})$ is zero: the next map $H_2(D; \mathbb{Z}) \rightarrow H_2(D, S; \mathbb{Z})$ is the non-singular hyperbolic intersection map. Also $H_1(S; \mathbb{Z}) \rightarrow H_1(D; \mathbb{Z})$ is the isomorphism in (17). Mayer–Vietoris, together with the isomorphism just proved for J , therefore identifies $H_2(D \cup_S J; \mathbb{Z})$ with $H_2(D; \mathbb{Z})$. All products are unchanged, since their representatives lie in the interior of D . Finally, the surjectivity of $\pi_1(S) \rightarrow H$ and van Kampen show that $D \cup_S J$ is simply connected. $\square$

Let $|v|$ denote Coxeter word length. Lemma 3.1 identifies the finite chamber union

$$U_L = \bigcup_{|v| \leq L} D_v$$

with an iterated boundary connected sum, with the natural markings on its boundary factors. Fill every noncentral boundary factor by the corresponding J_v of Lemma 5.1. The resulting manifold U_L^+ has boundary S and admits the description

$$U_L^+ \cong D_e \# \#_{0 < |v| \leq L} (D_v \cup_S J_v). \quad (19)$$

To see this description, retain the connecting three-balls in the boundary-sum construction. Gluing a punctured filling to a noncentral factor closes that factor away from its connecting ball and turns

the connecting boundary neck into an interior connected-sum neck. Repeating this one factor at a time gives (19).

It follows that $\pi_1(U_L^+) = H$, with the central boundary marking. On $\pi_1(U_L)$ the inclusion is the projection of the free product of the chamber groups onto its central factor. The universal cover of U_L^+ has reduced homology only in degree two, freely based over $\mathbb{Z}H$ by all the chamber sphere systems, including stabilizations. Its intersection form is their orthogonal hyperbolic sum. These facts also follow directly by lifting the connected-sum necks in (19); all the closed noncentral factors are simply connected and have precisely the middle classes computed above.

For an integer R , replace every standard neighborhood in $\mathcal{B}$ whose label belongs to a chamber $|v| \leq R$ by a four-ball. Call this operation *peeling*. The common boundary is a standard S^3 . There is a continuous pinch map from each old neighborhood to the new ball, fixed on that boundary, and these maps combine with the identity outside to a global pinch map. Both pieces and the common boundary are simply connected, so peeling preserves the fundamental group. The relative homology sequence deletes exactly the hyperbolic pair carried by that neighborhood and introduces no third homology.

We first perform this operation in the unbounded Y , obtaining Y_R . For L large enough to contain the selected neighborhoods and all their marking homotopies, it also defines a compact manifold

$$Z = Z_{R,L}$$

by peeling U_L^+ . The order of filling and peeling is immaterial because the standard neighborhoods are in the interior. The included homotopies identify the removed summands with exactly the chamber pairs labelled by $|v| \leq R$. We have therefore proved

$$\partial Z = S, \quad \pi_1 Z = H, \quad \tilde{H}_i(\tilde{Z}; \mathbb{Z}) = 0 \quad (i \neq 2), \quad (20)$$

and $H_2(\tilde{Z}; \mathbb{Z})$ is freely based by the projected chamber pairs with $R < |v| \leq L$, with their orthogonal hyperbolic form.

5.2 Uniform finite preparation

A compact object is *carried within b galleries of v* if its image is contained in chambers whose labels are within word distance b of v . Replacement balls are assigned the carriers of their old standard neighborhoods. We use this terminology for sphere maps, paths, homotopies and grope systems, including all their caps. A uniform propagation bound is one value of b valid for the whole labelled family. It is stronger than a bound after an unspecified permutation of the labels.

Lemma 5.2 (Finite preparation). *Fix a positive integer h . In Y_R , the projected pairs of $\mathcal{A}$ labelled by $|v| > R$ admit the following preparation, with propagation bounded independently of R :*

1. *The two halves are represented by framed immersed spheres a_i, b_i with $a_i \cap b_j$ consisting of the designated single point for $i = j$ and empty for $i \neq j$.*
2. *The remaining intersections among the a_i are paired by framed Whitney discs. Each Whitney disc is replaced by a properly immersed capped grope of height at least h with the same framed attaching region.*
3. *These gropes have the Clifford dual capped surfaces used in [17, Definition 3.1 and Lemma 4.1]. In particular, their caps miss the entire dual capped surfaces, and the bodies of the latter lie in a boundary collar of the complement of the a_i .*

The construction uses no complementary filling J_v . Beyond a fixed gallery distance from the peeling region, it can be carried out separately inside individual chambers.

Proof. First project the outer immersed systems through the pinch and put them in general position. The deleted and retained blocks were orthogonal, since their classes agree with the correspondingly labelled standard pairs. The retained forms and quadratic self-intersection data are therefore still hyperbolic over $\mathbb{Z}\pi_1(Y_R)$. Choose framed immersed representatives and cancel the algebraic intersection pairs. The framing corrections and, where necessary, local cusp changes can be chosen from finitely many local drawings. The group in question is a free product of free chamber groups, hence is torsion-free; there is no order-two ambiguity in these quadratic data.

We justify carefully the uniformity of the Whitney discs required at this and subsequent finite stages. The initial lifted data have finitely many types under a cocompact group of translations of Y . A gallery ball of fixed radius contains a uniformly bounded number of chambers and of pieces of those data. There are thus finitely many local templates, even after specifying which of the standard neighborhoods meeting that ball have been peeled. Paths on participating spheres are taken inside their bounded carriers. Common long whiskers conjugate an intersection label but need not be included in the geometric Whitney circuit.

If a circuit enters a replacement ball, move its portions there to chosen arcs on the boundary three-sphere. This gives a circuit in the unpeeled space. Nullity is unchanged, since both the deleted punctured $S^2 \times S^2$ and the replacement ball are simply connected and have simply connected boundary. For every null circuit among the finitely many resulting templates, choose one compact filling in Y . There is a finite maximum of their propagation bounds. Translate the chosen filling to the location of the circuit and apply the actual global pinch map. This constructs a filling in Y_R .

A filling may encounter additional peeled neighborhoods. No further choice of fillings is then needed: the global pinch is already defined on them, and each such neighborhood has the same bounded carrier. Consequently this procedure is uniform in R . It is a finite-template argument for the specified circuits, not a bound for arbitrary null loops. Repeating a fixed finite number of stages preserves the argument: enlarge the carrier radius and refine the finite template list at each stage.

Now apply the geometric Casson lemma to the two halves of the sphere system, removing the paired mixed intersections by regular homotopies. There is no iterative repair cascade here. Use the local construction of [17, Lemma 3.3] with the two surface arguments taken to be the entire possibly disconnected families. Preselect all framed pairing Whitney discs with disjoint boundaries; their interiors miss the other Whitney boundary arcs by $2 + 1 < 4$ general position. First push the first family off all these interiors by fingers along arcs in the Whitney discs. Choose the arcs away from their finitely many intersections with other Whitney discs and the second family. The finger supports then leave the other Whitney discs and attaching boundaries fixed. Next Whitney-move the second family using the same discs. The new intersections in each phase are only within that phase's family; intersections between Whitney interiors cause only allowed second-family intersections. Thus the whole batch stays in the original bounded Whitney neighborhoods. No later disc is transported or chosen again through an expanding sequence of carriers. Local finiteness permits these bounded batches on the unbounded Y_R . This produces the geometric duality in (1), while permitting self-intersections within either half. The remaining double points of the first half admit framed Whitney discs. Tube their interiors off the first-half spheres using the geometric duals. All the Whitney circuits involved are the null circuits just discussed.

For completeness, we verify the algebraic input to the grope preparation in the complement M_0 of small plumbed neighborhoods of the a_i . The punctured geometric dual b_i bounds an a_i -meridian

in M_0 . Since these meridians normally generate the kernel, inclusion induces $\pi_1 M_0 \cong \pi_1 Y_R$. Thus the following calculation retains all intersection labels in the required complement group ring.

At one point of every paired double point take its Clifford torus. Cap its two basis curves by standard meridional discs and tube each meridional disc off its incident a_i -sheet into a punctured parallel of b_i , with the product framing. Denote these caps by C_ℓ , and let $i(\ell)$ be the index of the dual used. Their equivariant intersection calculation is the Clifford-cap calculation in the proof of [17, Section 5]. The local meridional pieces have disjoint interiors. Choose the thin tubing tracks away from the finitely many other designated dual points; local changes of tracks introduce only oppositely signed points with the same group label. The remaining intersections are those between the corresponding based parallels of $b_{i(\ell)}$. Hence their sums are obtained from the pairings of the b_i by the basing units and orientation signs. Likewise, a cap's self-intersection sum is that of its framed b_i parallel, together with cancelling local terms. Since the mixed Casson regular homotopies preserved $\lambda(b_i, b_j) = 0$ and $\mu(b_i) = 0$, this proves

$$\lambda(C_\ell, C_m) = 0, \quad \mu(C_\ell) = 0 \quad \text{for every } \ell, m,$$

including distinct parallel caps using the same dual sphere.

The Clifford bodies lie in a boundary collar of M_0 . Following [17, Section 5], contract temporary parallel copies and push the Whitney discs off those contractions before using them as geometric dual spheres. At the remaining intersections with Whitney discs, tube the Clifford caps into parallel copies of these temporary duals. Lemma 4.1 of [17] now applies with its full cap-pairing hypothesis. Its two separation batches and height raising provide the prepared Clifford copies, whose entire capped surfaces are geometrically dual in the sense of [17, Definition 3.1]. These prepared copies, including all their caps, are the protected surfaces used from this point onward. The preparation in [17, Lemma 4.1] produces height-two Whitney gropes, separates their caps from the protected Clifford caps, and raises the height to h . The collar levels and the full dual-cap disjointness are part of this preparation, not an assertion made after the caps have been changed. There are exactly two cap-family separation batches before height raising. Each uses the same no-new-mixed-pairs invariant, with its protected bodies at fixed collar levels.

Each of the operations before height raising involves only a fixed number of kinds of tubing, push-off, Whitney-disc choice and regular homotopy. Their simultaneous use does not require traversing the whole list of sphere components. At each fixed stage, bounded carriers and finite drawings bound the number of other carriers meeting a given one. Divide the operations into finitely many classes with disjoint enlarged carriers and perform one class at a time. This gives a number of local stages independent of R . New null circuits are handled by the finite-template procedure above.

For height raising there is also an explicit size estimate: [9, Section 2.8.3], in the January 2013 numbering, bounds the increase of component-image diameter by a factor of seven at each step. A fixed requested height therefore has a fixed propagation bound. One can apply this estimate to the gallery map into the Davis complex: extend that map over each replacement ball by coning its boundary image inside a metric ball of uniformly bounded radius. This is possible by CAT(0) convexity. A point in a chamber is mapped within a fixed distance of its label, and properness of the Coxeter action turns a fixed metric bound into a fixed word-distance bound. The construction uses parallel transverse gropes and neighborhood operations; it preserves the protected Clifford capped surfaces. All these operations can be made locally finite in the unbounded development.

Finally, outside a fixed buffer of the peeling region the input is the original system in an individual chamber. Its intersection cancellations hold already over that chamber's group ring. The same preparation, including the null discs, can consequently be chosen in that chamber. Only

a bounded buffer of the switch between the peeled and unpeeled data requires the preceding translated templates. This proves the last assertion and permits later truncation without cutting off an unfinished preparation. $\square$

5.3 Directions at infinity

Let Σ_C be the ordinary Davis complex and let o be its vertex labelled by $e \in C$. The gallery map of Lemma 3.4 sends the inner part of D_v to v , interpolating across the dual collars. The link of o is S . A small metric sphere of radius t about o is consequently identified with S by radial scaling in its conical star. Write ρ for radial projection onto this sphere, defined outside the open ball of radius t .

Lemma 5.3 (Small directions and the central marking). *Fix a propagation bound b . For objects carried within b galleries of labels v with $|v| \rightarrow \infty$, their direction images in S have diameter tending uniformly to zero. This assertion continues to hold after peeling for the whole region of bounded propagation around the outer systems, if R is sufficiently large.*

Let $\beta : S \rightarrow BH$ be the central boundary marking. On that region, the homomorphism induced by the direction map followed by β agrees, up to the common change of base path, with inclusion into Z followed by its classifying map. In particular, every loop in an image system whose entire direction image lies in a contractible patch of S is nullhomotopic in Z .

Proof. Here is the radial estimate that supplies the first assertion. If $x, y \in \Sigma_C$ have distance at least $s > t$ from o , put them first on the same metric sphere, of radius the smaller of their two radii. The distance between the resulting points is at most $2d(x, y)$, by the triangle inequality and $|d(o, x) - d(o, y)| \leq d(x, y)$. CAT(0) comparison on the two radial segments then gives

$$d(\rho(x), \rho(y)) \leq \frac{2t}{s} d(x, y). \quad (21)$$

The gallery map takes each chamber into a set of uniformly bounded diameter. Thus a bounded gallery carrier has bounded image diameter. Properness of the Davis complex, together with its cocompact Coxeter action, makes its distance from o tend to infinity with the word length of its center. Equation (21) applies.

We only need the direction map outside the central core. Consider a peeled standard neighborhood that meets an outer object or any of its fixed enlarged carriers. Both the object and the whole standard neighborhood have bounded propagation. The label of this neighborhood therefore has length at least R minus a fixed constant. Its entire boundary three-sphere, not merely the arcs encountered by the object, has direction image of arbitrarily small diameter for large R . Choose a slightly larger coordinate ball of S containing this image. Contracting that ball extends the boundary map across the replacement four-ball, with image still in the chosen ball. These extensions agree with the unchanged direction map on all their boundaries.

The near-central peeled neighborhoods need not admit such small extensions. They are disjoint from all the carriers under consideration once R is large, and no direction map on them is used. This proves the asserted far-out extension and its uniform smallness.

Before peeling, the composition with β extends across the central chamber by its given map to BH . Its restriction to the central boundary agrees with that map: this is the boundary-direction identification in Lemma 3.4. In a noncentral chamber the gallery map collapses its inner part to the noncentral vertex, so every loop of its chamber group has constant direction there. The shelling description makes the chamber groups the free factors of the fundamental group of a finite

ball of chambers. The induced homomorphism is thus exactly the projection retaining the central factor. By (19) this is also the homomorphism induced by passage to U_L^+ .

Replacing a simply connected piece by a simply connected ball does not alter this calculation. Loops through the replacement can be moved to the boundary of the piece, and the direction extension agrees there with the old map. The comparison therefore holds on the required region in Z . If a whole image system maps into a contractible patch, every one of its loops, including loops passing through arbitrarily many intersections, has trivial image under β . Since BH is a classifying space and $\pi_1 Z = H$, these loops are nullhomotopic in Z . $\square$

5.4 Upper grope separation

We state explicitly how the controlled data supply the input used after the good-group step of the disc embedding proof. The units to be separated are the upper capped subgroves attached to individual tips of the retained stages. Separating only whole Whitney groves would not suffice to make the future cap discs mutually disjoint.

Lemma 5.4 (Null upper replacements). *Suppose a finite prepared Whitney-grope system as in Lemma 5.2 has at least eight surface stages. Retain its first five stages. Suppose the upper capped subgroves, of height at least three, have uniformly small direction images in the region of Lemma 5.3. If their diameters are sufficiently small, depending only on a fixed finite coordinate cover of S , the upper subgroves can be replaced by mutually disjoint framed immersed cap discs. Their self-intersection loops are nullhomotopic in Z . The retained stages and the entire protected Clifford capped surfaces remain unchanged.*

Proof. We first specify the relative setting. Remove a sufficiently small regular neighborhood of the five retained stages, truncating each upper attaching collar at its boundary. The upper components are now properly immersed disk-like capped groves of height at least three. By the duality condition in Lemma 5.2, their entire images miss the protected Clifford capped surfaces. Choose a neighborhood of the upper system disjoint from those surfaces and from the retained stages except at the prescribed attaching regions. All the following modifications are neighborhood operations in this relative manifold. They therefore leave the lower stages, attaching framings, and whole Clifford systems unchanged, including the unused boundary-collar levels needed later.

We use the following precise content of Freedman–Quinn’s controlled separation lemma [9, Lemma 2.10.1]. For a map from a four-manifold to a locally compact metric space, a properly immersed union of disk-like capped groves of height at least three, with sufficiently small component images, admits new caps for the *same bodies* such that distinct components are disjoint and their images remain small. Its compact-case proof separates finitely many cover collections before separating the members of each collection [9, Section 2.10.2]. We retain this two-phase construction because we need a bound on gallery propagation only in its first phase.

Choose compact sets $K_1, \dots, K_q$ covering S , each contained in an open coordinate ball V_j . Choose $\epsilon > 0$ so that the closed ϵ -neighborhood of every K_j lies in V_j . Assign each upper component to a K_j met by its direction image. If every initial image has diameter less than δ , that component lies within distance δ of its assigned K_j .

For the first phase, regard all components assigned to K_j as one collection. The construction in [9, Section 2.10.2] raises height by one, contracts the added stages of the first collection and pushes the later collections off, then continues through the collections in order. The original bodies are retained. Its component-diameter estimate is $3^{q-1}7^2\delta$. A point of each retained body stays within δ of K_j , so the entire resulting j th collection lies in the $3^{q+3}\delta$ -neighborhood of K_j . Choose δ with

$3^{q+3}\delta < \epsilon$, leaving room for a small regular-neighborhood thickening. The number of height-raising and contraction passes depends only on q . The same size estimates apply to the gallery map into Σ_C , extended over the replacement balls as in Lemma 5.2. Since each original body is retained, its image anchors the resulting component within a bounded metric distance of the original label. Properness of the Coxeter action converts this into a gallery bound independent of the number of components and of R .

The collections now have pairwise disjoint images. Choose disjoint regular neighborhoods of their entire images, with the j th neighborhood still mapping into V_j . Inside each such neighborhood, apply the collection-separation procedure again, this time taking each individual upper component as a collection. There may be arbitrarily many of them. The height is raised once more inside that neighborhood before contracting the added stages, so this second phase again retains the original bodies. No estimate depending on their number is needed: all operations stay in the already chosen neighborhood. The resulting upper components have pairwise disjoint neighborhoods. Totally contract each in its own neighborhood to obtain the claimed mutually disjoint framed immersed cap discs.

Every loop in a resulting cap image lies in its color neighborhood, whose entire direction image is contained in the contractible V_j . Lemma 5.3 therefore makes that loop nullhomotopic in Z . The second phase need not bound the gallery diameter of an individual cap. It preserves instead the containment of its whole image in a direction patch, which is exactly the condition needed for nullity. $\square$

5.5 Choice of parameters and the tower construction

Fix the coordinate cover in Lemma 5.4. Request height eight in Lemma 5.2. Its uniform propagation bound, together with the bounds for the standard neighborhoods and marking homotopies, controls the entire preparation near the peeling boundary. The first phase of upper separation adds only the fixed height extension and q collection passes described above. Let b bound the gallery propagation of all these operations, independently of R . Choose R so large that the direction estimates of Lemma 5.3, including the replacement-ball extensions, meet the chosen cover margins for every such carrier.

We now truncate the prepared family, before carrying out the collection-separation operations. Choose L large enough to contain every peeled standard neighborhood and its marking homotopies, and the whole b -buffer around the transition between peeled and unpeeled data. Beyond that buffer, Lemma 5.2 constructs the sphere and Whitney-grope data entirely inside their own chambers, using their chamber group-ring pairings. Retain precisely the planes whose labels satisfy $R < |v| \leq L$, together with their complete preparations. The preparations at the outer boundary lie in chamber interiors, so none has a partner or an attaching region outside U_L . This gives a finite family in the peeled copy of U_L , before any complementary filling is used.

Cap its noncentral boundary factors to form $Z = Z_{R,L}$. Apply the first phase of Lemma 5.4 to this finite family. All of its operations can be made in arbitrarily small neighborhoods of the retained upper system inside that peeled region; the bound b supplies the direction margins independently of the truncation. Apply the second phase inside the resulting disjoint color neighborhoods. Neither phase requires an enlargement of L or uses the complementary fillings. The second phase supplies the immersed null-loop caps without a uniform gallery bound on their individual images.

Write

$$M' = Z \setminus \bigcup_i \text{int } \nu(a_i)$$

for the complement used in the Whitney construction. The spheres a_i may still be immersed; the notation denotes their standard plumbed neighborhoods. The prepared geometric dual spheres imply

$$\pi_1(M') \xrightarrow{\cong} \pi_1(Z). \quad (22)$$

Indeed the kernel of the inclusion map is normally generated by meridians of the a_i , and the punctured dual spheres bound these meridians in the complement. General position supplies surjectivity. This is the usual π_1 -negligibility provided by geometric duality.

The new caps therefore have null double-point loops in M' , not just in Z . They are mutually disjoint, although each may have self-intersections. Direction control has now supplied the needed fundamental-group condition. The remaining constructions require that condition and the protected geometric duals, but no bound on the diameter of the null homotopies. Symmetrically contract the fifth retained surface stage. The result is a height-four capped Whitney grope system, still with mutually disjoint caps. Its cap self-intersections occur in algebraically cancelling pairs, and their loops are parallel copies of the previous null loops. This is exactly the intermediate input in the proof of [17, Lemma 4.2], after that proof has used goodness. The retained Clifford capped surfaces are still geometrically dual in the stronger sense of [17, Definition 3.1].

We explain why the remainder of that proof applies without any assumption on H . Cap the null double-point loops by immersed discs in M' . The bodies of the Clifford capped surfaces lie at the protected boundary-collar levels. The attaching loops of the new discs miss that collar, so the null homotopies can be chosen to miss the Clifford *bodies* by pushing them out of the collar. They may meet the Clifford caps and may meet the Whitney gropes arbitrarily. Correct their boundary framings, push their intersections with the gropes down, and tube into geometrically dual spheres obtained by contracting suitable parallel copies of the Clifford systems. These operations produce the tower caps. At the next collar level, contract the remaining Clifford capped surfaces and push the tower caps off those contractions. The resulting spheres are geometric duals to the one-storey capped towers. These are the successive operations in the remainder of the proof of [17, Lemma 4.2]; no subsequent step uses goodness.

One-storey capped towers with four surface stages contain the embedded discs required for the Whitney moves, with the same framed attaching regions. Applying the final steps of [17, Section 5], first move the original sphere duals off the towers by tubing into the tower duals. Then use the embedded Whitney discs to remove the intersections of the a_i . We obtain disjoint framed embedded representatives of their original Lagrangian classes, together with geometric transverse spheres. The changes of upper caps do not change the retained Whitney attaching data or the corrected sphere classes. In particular, the symmetric-contraction independence of cap choices in [17, Lemma 3.4] applies at the retained stages.

There is deliberately no assertion that the null homotopies, the tower caps, or the final embedded Whitney discs have small gallery diameter. From (22) onwards they may range throughout Z . Only their existence, their relative attaching data, and the geometric duals are needed.

Proposition 5.5 (Removal of the remaining planes). *The compact manifold Z admits interior surgeries producing a compact oriented topological four-manifold V with boundary S , fundamental group H , and contractible universal cover. The boundary map to BH is the original central marking.*

Proof. Perform surgery on the disjoint framed Lagrangian spheres just constructed. Their geometric duals kill the meridians in their complement, so the surgeries preserve the fundamental group. All changes are in the interior; the boundary and its marking are unchanged.

For clarity, let there be m remaining planes and let W be the five-dimensional surgery trace. With regular $\mathbb{Z}H$ coefficients, the relative chain complex of (W, Z) is concentrated in degree three, with module $(\mathbb{Z}H)^m$. Its boundary map to $H_2(Z; \mathbb{Z}H)$ sends the basis to the Lagrangian basis. By (20) this is a split injection. Consequently

$$H_3(W; \mathbb{Z}H) = 0, \quad H_2(W; \mathbb{Z}H) \cong (\mathbb{Z}H)^m,$$

the latter module represented by the dual half, and the other positive-degree homology groups vanish. Read the same trace from V . Its relative chain complex is concentrated in degree two, again with module $(\mathbb{Z}H)^m$. The map

$$H_2(W; \mathbb{Z}H) \longrightarrow H_2(W, V; \mathbb{Z}H)$$

is the pairing of the surviving dual classes with the surgered Lagrangian, hence is an isomorphism. The sequence of (W, V) therefore gives $H_i(V; \mathbb{Z}H) = 0$ for every $i > 0$.

The universal cover of V is simply connected and acyclic. A topological manifold has CW homotopy type [15, Corollary 1]; in the boundary case one may first append an exterior open collar, which does not change homotopy type. The homological Whitehead theorem therefore makes this universal cover contractible [10, Corollary 4.33]. Thus V is a $K(H, 1)$ and is homotopy equivalent to the finite graph BH . Under this equivalence its boundary map is the retained central marking, as asserted. $\square$

The manifold V is the marked graph-type filling excluded by Proposition 2.1. This contradiction rules out the hypothetical closed aspherical topological four-manifold and completes the proof of Theorem 1.1.

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